

02451 Introduction to Machine Learning

Week 9: Bias/Variance and ensemble methods

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7th April

DTU Compute, Technical University of Denmark



Today

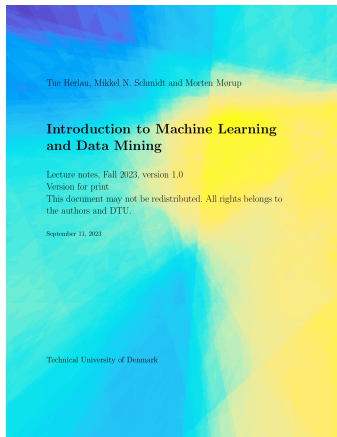
Feedback Groups of the day:

Anton Hammerich Abildstrøm, Lucinda Ani Ackland, Lau Gad Akrawi, Anna Kathrine Krogh Albrechtsen, Noor Al-Kashmiri, Lorenzo Antonini, Benjamín Kári Atlason, Matilde Bogana, Lucas Patricio Martin Campopiano, Alberte Bech Elholm, Isak Fazal, Hamza Hamdaoui, Enric Hernandez Minaya, Kristoffer Haaber Hoegh, August Johannes Hoier, Yun Jie Si Høj, Freja Cecilie Heister Jensen, Rasmus Ladegaard Jensen, Jan-Luca Laubner, Louisa Amalie Berg Lykkesfeldt, Gunwoo Na, Gustav Dalgård Oxenbøll, Margot Perrichon de La Rochebrochard, Oliver Petersen, Shourya Prakhara, Fie Karoline Gundgaard Schou, Louise Seedorff, Johannes Munk Sigsgaard, Jóhannes Sigvardsen, Viggo Starcke, Helene Log Staveland, Simon Strobel, Ahmad Mahmoud Yassin, Yutao Zhang.

Reading/homework material:

Chapter 14, 17

F24 Q8, P17.1



Lecture Schedule

- 1 Introduction, data, and visualization
3 February: C1, C2, C7
- 2 Summary statistics, similarity, and nearest neighbors
10 February: C4, C12
- 3 Computational linear algebra and PCA
17 February: C3
- 4 Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5 Bayesian inference and linear models
3 March: C5, C6, C8
- 6 Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7 Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8 Artificial Neural Networks and Optimization
24 March: C15
- 9 **Bias/Variance and ensemble methods**
7 April: C14, 17
- 10 K-means and hierarchical clustering
14 April: C18 (Assignment 7-9 due on 16 April)
- 11 Mixture models and density estimation
21 April: C19, C20
- 12 Representation learning
28 April: Material provided separately
- 13 Recap and discussion of the exam
5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Practicalities and announcements

- Midterm "test"
- Notes for exam

Recap Lecture 8

Generalized Linear Model (GLM)

Data: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$

Goal: $f_w: x \rightarrow y$

Model: $f_w(x) = h(w^T x)$
where h is a "simple" link func.

Cost: $d(f_w(x), y)$

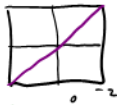
$w = \arg \min_w \sum_{i=1}^N d(f_w(x_i), y_i)$
Loss E(w)



Regression

$h(z) = z$

$d(f_w(x), y) = (f_w(x) - y)^2$



ReLU

$h(z) = \max\{0, z\}$



"Bounded" Reg.

$h(z) = \tanh(z)$

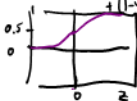
$d(f_w(x), y) = |f_w(x) - y|$



Binary Classification

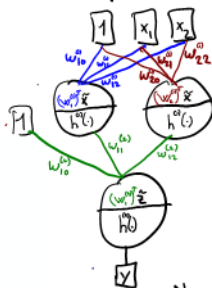
$h(z) = \frac{1}{1 + e^{-z}} = \sigma(z)$

$d(f_w(x), y) = y \cdot \log \sigma(w^T x) + (1 - y) \cdot \log (1 - \sigma(w^T x))$



Artificial Neural Network (ANN)

A layered composition of many simple nonlinear functions $\Rightarrow$ complex mapping from input to output.



$$\underline{w}^{(1)} = \begin{bmatrix} -(w_{10}^{(1)})^T \\ -(w_{20}^{(1)})^T \end{bmatrix} = \begin{bmatrix} w_{10}^{(1)} & w_{11}^{(1)} & w_{12}^{(1)} \\ w_{20}^{(1)} & w_{21}^{(1)} & w_{22}^{(1)} \end{bmatrix}$$

$$z = h^{(1)}(\underline{w}^{(1)} \tilde{x})$$

$$\underline{w}^{(2)} = [-w_{30}^{(2)}] = [w_{30}^{(2)} \ w_{31}^{(2)} \ w_{32}^{(2)}] \text{ (here a vector)}$$

$$f_w(x) = h^{(2)}(\underline{w}^{(2)} \tilde{z})$$

$$= h^{(2)}(\underline{w}^{(2)} [1 \ h^{(1)}(\underline{w}^{(1)} \tilde{x})]^T)$$

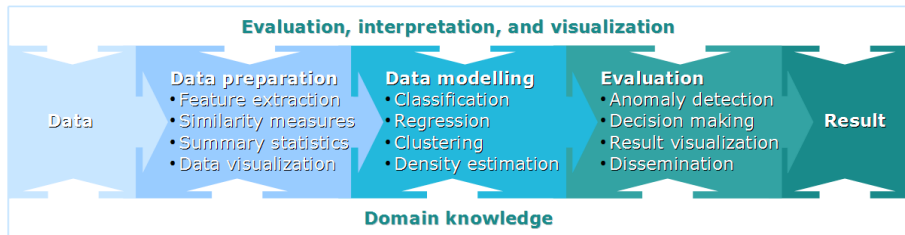
$$\underline{w}^* = \arg \min_{\underline{w}} \sum_{i=1}^N d(f_w(x_i), y_i)$$

Remarks

- Choice of #hidden units, activation functions $h(\cdot)$, #hidden layers, loss $\Rightarrow$ could use CV!
- No closed-form solution (except trivial case) $\Rightarrow$ Use gradient-based methods
- Multiple outputs possible! For regression combine MSE losses
- For multiclass problems use eg. softmax $p(y=k | f(x), \theta) = \frac{e^{f_k(x)}}{\sum_{c=1}^C e^{f_c(x)}}$

Alternative: multinomial regression parametrisation (saves one output)

Learning Objectives



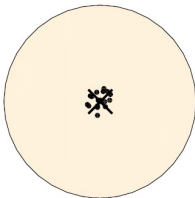
Learning Objectives

- Understand the Bias-Variance decomposition
- Understand and apply regularized least squares regression (i.e. ridge regression), logistic regression and neural networks.
- Explain the principle behind boosting and bagging and apply it to improve classifiers

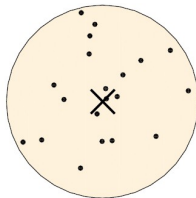
Bias/Variance decomposition and Regularization

What is bias and what is variance?

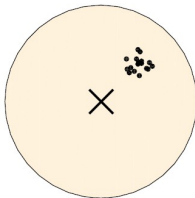
Low bias low variance



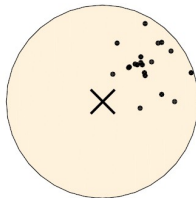
Low bias high variance



high bias low variance



High bias high variance



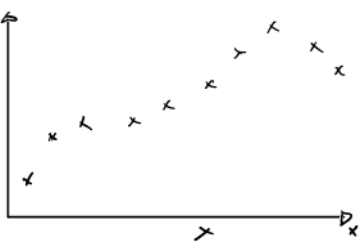
Recall: Least-squares regression

$$f(x; w) = w^T \tilde{x} \\ (= w_0 + w^T \hat{x})$$

$$y_i = f(x_i; w) + \varepsilon \\ \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$\hat{x} = [1 \ x \ x^2 \ \cos(x) \ x^3 \ x^4 \ \exp\{x\}]$$

$$w^* = \underset{w}{\operatorname{argmin}} \sum_{i=1}^N (y_i - f(x_i; w))^2$$



Regularized least squares

- Cost function from linear regression

$$E(\mathbf{w}) = \left\| \mathbf{y} - \tilde{\mathbf{X}} \mathbf{w} \right\|^2$$

- A parsimonious model can be obtained by **forcing** parameters towards zero.
- Problem: Columns of $\mathbf{X}$ sometimes have very different scale (i.e. require large/small values of $\mathbf{w}$); therefore, standardize $\mathbf{X}$:

$$\hat{X}_{ij} = \frac{X_{ij} - \mu_j}{\hat{s}_j}, \quad \mu_j = \frac{1}{N} \sum_{i=1}^N X_{ij}, \quad \hat{s}_j = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_{ij} - \mu_j)^2}$$

- Note $\hat{\mathbf{X}}$ contains no constant term.

$$\|\mathbf{w}\|_2^2 = w_1^2 + w_2^2 + \dots + w_M^2$$

- Introduce regularization term $\lambda \|\mathbf{w}\|^2$ to penalize large weights:

$$E_\lambda(\mathbf{w}, w_0) = \sum_{i=1}^N (y_i - w_0 - \hat{\mathbf{x}}_i^\top \mathbf{w})^2 + \lambda \|\mathbf{w}\|^2 = \left\| \mathbf{y} - w_0 \mathbf{1} - \hat{\mathbf{X}} \mathbf{w} \right\|^2 + \lambda \|\mathbf{w}\|^2$$

- We can solve for w_0 and $\mathbf{w}$:

$$\begin{aligned} \frac{dE_\lambda}{dw_0} &= \sum_{i=1}^N -2(y_i - w_0 - \hat{\mathbf{x}}_i^\top \mathbf{w}) = -2N\mathbb{E}[y] - 2Nw_0 - N \left(\frac{1}{N} \sum_{i=1}^N \hat{\mathbf{x}}_i^\top \right) \mathbf{w} \\ &\Rightarrow w_0 = \mathbb{E}[y] \end{aligned}$$

- With $\hat{y}_i = y_i - \mathbb{E}[y]$

$$E_\lambda = \left\| \hat{\mathbf{y}} - \hat{\mathbf{X}} \mathbf{w} \right\|^2 + \lambda \|\mathbf{w}\|^2$$

- Setting the derivative wrt. $\mathbf{w}$ equal to zero and solving for $\mathbf{w}$ yields

$$\mathbf{w}^* = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}} + \lambda \mathbf{I})^{-1} (\hat{\mathbf{X}}^\top \hat{\mathbf{y}})$$

Selecting λ

- Suppose we have a 1D problem

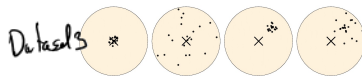
$$w^* = (\hat{X}^T \hat{X} + \lambda I)^{-1} (\hat{X}^T \hat{y}) \propto \frac{\hat{X}^T y}{\hat{X}^T \hat{X} + \lambda}$$

- So if $\lambda = 0$ then no effect, else if $\lambda \rightarrow \infty$ then $w^* \rightarrow 0$
- λ controls complexity of model. Select λ using cross-validation

How does different values of λ (vertical) affect the bias/variance of learned function (red lines)

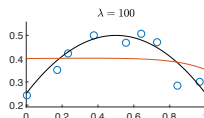
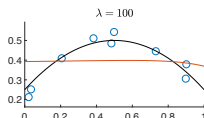
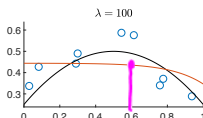
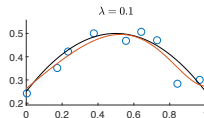
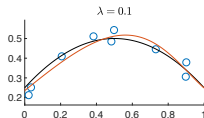
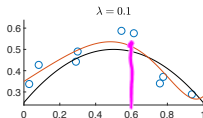
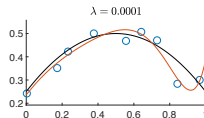
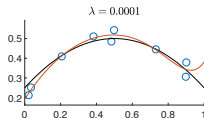
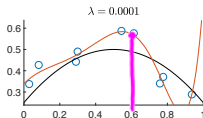
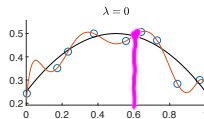
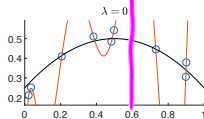
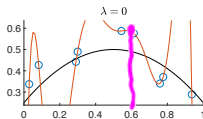
Dataset 1

Dataset 2



High var
(Low bias)

High bias
Low var



Bias-variance decomposition

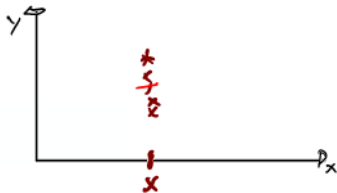
$$\begin{aligned}
 \mathbb{E}_{\mathcal{D}}[E^{gen}] &= \mathbb{E}_{\mathcal{D},(x,y)} [(y - f_{\mathcal{D}}(x))^2] \\
 &= \mathbb{E}_{\mathcal{D}} \left\{ \mathbb{E}_{(x,y)} \left\{ (y - f_{\mathcal{D}}(x))^2 \right\} \right\} \\
 &= \iint (y - f_{\mathcal{D}}(x))^2 \underbrace{p(x,y)p(\mathcal{D})}_{p(y|x)p(x)} dx dy d\mathcal{D}
 \end{aligned}$$

We first consider x fixed

$$\begin{aligned}
 &\mathbb{E}_{\mathcal{D},y|x} [(y - f_{\mathcal{D}}(x))^2] \\
 &= \mathbb{E}_{\mathcal{D},y|x} [(y - \bar{y}(x) + \bar{y}(x) - f_{\mathcal{D}}(x))^2] \\
 &= \mathbb{E}_{y|x} [(y - \bar{y}(x))^2] + \mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - f_{\mathcal{D}}(x))^2] + 2\mathbb{E}_{\mathcal{D},y|x} [(y - \bar{y}(x))(\bar{y}(x) - f_{\mathcal{D}}(x))]
 \end{aligned}$$

$$\bar{y}(x) = \mathbb{E}_{y|x}[y]$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$



Bias-variance decomposition

$$\mathbb{E}_{\mathcal{D}}[E^{gen}] = \mathbb{E}_{\mathcal{D}, (x, y)} [(y - f_{\mathcal{D}}(x))^2]$$

We first consider x fixed

$$\mathbb{E}_{\mathcal{D}, y|x} [(y - f_{\mathcal{D}}(x))^2] \qquad \bar{y}(x) = \mathbb{E}_{y|x}[y]$$

$$= \mathbb{E}_{\mathcal{D}, y|x} [(y - \bar{y}(x) + \bar{y}(x) - f_{\mathcal{D}}(x))^2]$$

$$= \mathbb{E}_{y|x} [(y - \bar{y}(x))^2] + \mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - f_{\mathcal{D}}(x))^2] + \cancel{2\mathbb{E}_{\mathcal{D}, y|x} [(y - \bar{y}(x))(\bar{y}(x) - f_{\mathcal{D}}(x))]}$$

The Bias-Variance decomposition

$$\bar{y}(x) = \mathbb{E}_{y|x} \{y\}$$

$$\mathbb{E}_{\mathcal{D}, y|x} [(y - f_{\mathcal{D}}(x))^2] = \mathbb{E}_{y|x} [(y - \bar{y}(x))^2] + \mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - f_{\mathcal{D}}(x))^2]$$

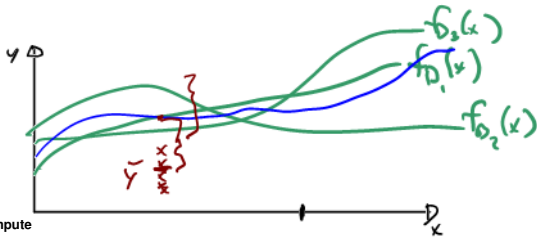
$$\mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - f_{\mathcal{D}}(x))^2]$$

2.

$$\bar{f}(x) = \mathbb{E}_{\mathcal{D}} [f_{\mathcal{D}}(x)]$$

$$= \mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - \bar{f}(x) + \bar{f}(x) - f_{\mathcal{D}}(x))^2]$$

$$= \mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - \bar{f}(x))^2] + \mathbb{E}_{\mathcal{D}} [(\bar{f}(x) - f_{\mathcal{D}}(x))^2] + 2\mathbb{E}_{\mathcal{D}} [(\bar{y}(x) - \bar{f}(x))(\bar{f}(x) - f_{\mathcal{D}}(x))]$$



The Bias-Variance decomposition

$$\mathbb{E}_{\mathcal{D}, y|x} \left[(y - f_{\mathcal{D}}(\mathbf{x}))^2 \right] = \mathbb{E}_{y|x} \left[(y - \bar{y}(\mathbf{x}))^2 \right] + \mathbb{E}_{\mathcal{D}} \left[(\bar{y}(\mathbf{x}) - f_{\mathcal{D}}(\mathbf{x}))^2 \right]$$

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \left[(\bar{y}(\mathbf{x}) - f_{\mathcal{D}}(\mathbf{x}))^2 \right] & \bar{f}(\mathbf{x}) &= \mathbb{E}_{\mathcal{D}} [f_{\mathcal{D}}(\mathbf{x})] \\ &= \mathbb{E}_{\mathcal{D}} \left[(\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}) + \bar{f}(\mathbf{x}) - f_{\mathcal{D}}(\mathbf{x}))^2 \right] \\ &= \mathbb{E}_{\mathcal{D}} \left[(\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2 \right] + \mathbb{E}_{\mathcal{D}} \left[(\bar{f}(\mathbf{x}) - f_{\mathcal{D}}(\mathbf{x}))^2 \right] + \cancel{2\mathbb{E}_{\mathcal{D}} \left[(\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}))(\bar{f}(\mathbf{x}) - f_{\mathcal{D}}(\mathbf{x})) \right]} \end{aligned}$$

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}, y|x} \left[(y - f_{\mathcal{D}}(\mathbf{x}))^2 \right] \\ &= \mathbb{E}_{y|x} \left[(y - \bar{y}(\mathbf{x}))^2 \right] + (\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2 + \mathbb{E}_{\mathcal{D}} \left[(\bar{f}(\mathbf{x}) - f_{\mathcal{D}}(\mathbf{x}))^2 \right] \\ &= \text{Var}_{y|x} [y] + \underbrace{(\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2}_{\text{Bias}^2} + \underbrace{\text{Var}_{\mathcal{D}} [f_{\mathcal{D}}(\mathbf{x})]}_{\text{Variance}} \end{aligned}$$

The Bias-Variance decomposition

(now consider all x)

$$\begin{aligned}\mathbb{E}_{\mathcal{D}}[E^{gen}] &= \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}, y | \mathbf{x}} \left[(y - f_{\mathcal{D}}(\mathbf{x}))^2 \right] \right] \\ &= \mathbb{E}_{\mathbf{x}} \left[\text{Var}_{y | \mathbf{x}}[y] + (\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2 + \text{Var}_{\mathcal{D}}[f_{\mathcal{D}}(\mathbf{x})] \right]\end{aligned}$$

The Bias-Variance decomposition

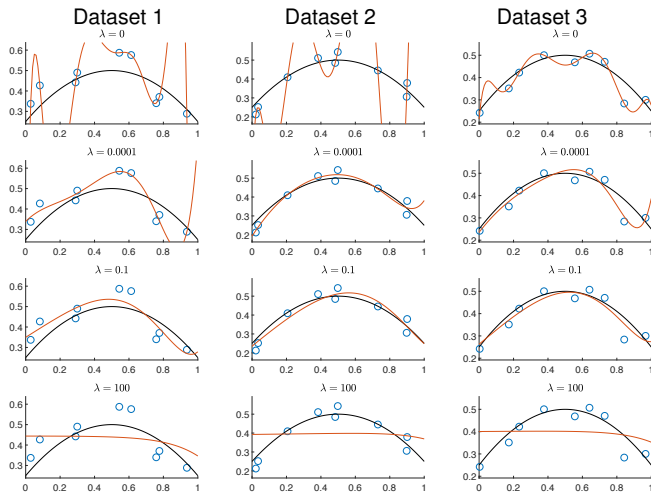
$$\mathbb{E}_{\mathcal{D}}[E^{gen}] = \mathbb{E}_{\mathbf{x}} \left[\text{Var}_{y|\mathbf{x}}[y] + (\bar{y}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2 + \text{Var}_{\mathcal{D}}[f_{\mathcal{D}}(\mathbf{x})] \right]$$

The first term does not depend at all upon our choice of model but simply represents the intrinsic difficulty of the problem. We cannot make this term any larger or smaller by selecting one model over another.

The second term is the **bias** term. It tells us how much the average values of models trained on different training datasets differ compared to the true mean of the data.

The third term is the **variance** term. It tells us how much the model wiggles when trained on different sets of training data. That is, when you train the models on N different (random) sets of training data and the models (the prediction curves) are nearly the same this term is small.

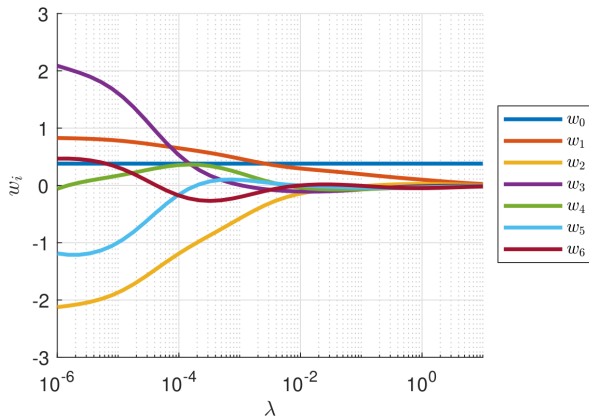
The bias variance decomposition



By regularization we can trade-off bias and variance, in particular, we can hope to substantially reduce variance without introducing too much bias!

Parameters w^* as function of λ

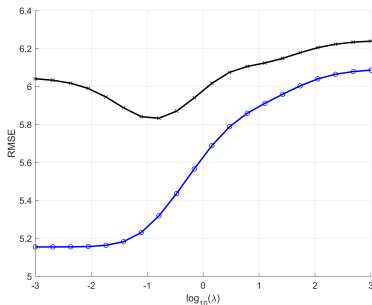
$$E_\lambda(\mathbf{w}) = \sum_{i=1}^N (\hat{y}_i - w_0 - \hat{\mathbf{x}}_i^\top \mathbf{w})^2 + \lambda \|\mathbf{w}\|^2$$



14:10

$$\|w\|_2^2 = w^T w$$

Quiz 1, Bias-variance



Using 54 observations of a dataset about Basketball, we would like to predict the average points scored per game (y) based on the four features. For this purpose we consider regularized least squares regression which minimizes with respect to w the following cost function:

$$E(w) = \sum_n (y_n - [1 \ x_{n1} \ x_{n2} \ x_{n3} \ x_{n4}]w)^2 + \lambda w^T w,$$

We consider 20 different values of λ and use leave-

one-out cross-validation to estimate the performance (measured by mean-squared error) of each of these different values of λ and plot the result in the figure. For the value of $\lambda = 0.6952$ the following model is identified:

$$f(x) = 2.76 - 0.37x_1 + 0.01x_2 + 7.67x_3 + 7.67x_4.$$

Which one of the following statements is correct?

- ☒ A. In the figure the blue curve with circles corresponds to the training error whereas the black curve with crosses corresponds to the test error.
- ☐ B. According to the model defined for $\lambda = 0.6952$ increasing a player's height x_1 will increase his average points scored per game.
- ☐ C. There is no optimal way of choosing λ since increasing λ reduces the variance but increases the bias.
- ☐ D. As we increase λ the 2-norm of the weight vector w will also increase.
- ☐ E. Don't know.

The correct answer is *A*: The blue curve monotonically increases with λ reflecting a worse fit to the training set as we increase λ using regularization we can reduce the variance by introducing bias and the black curve indicates that an optimal tradeoff at around $10^{-0.8}$ as reflected by the test error indicated in the black curve being minimal. As we increase λ we will

penalize the weights according to the squared 2-norm more and more and thus the 2-norm will be reduced. Finally, according to the fitted model we observe that the coefficient in front of x_1 (Height) is negative thus indicating that an increase in height will reduce the models prediction of average points scored per game.

Weight-based (e.g. L2) regularization as a general tool

- Generalized linear models (GLM)
 - Regularized linear regression (Ridge)
 - Regularized logistic regression
 - Multinomial regression
- Non-linear models
 - Neural networks
 - Kernel methods (not covered in this course)

$$\|w\|_2^2$$

$$\|w\|_1^2$$

$$\|w\|_0$$

Ensemble methods

So far....

Methods

- Trees
- Linear models
 - Lin. reg.
 - Log. reg.
- Nearest neighbor
- (Naïve) Bayes Class.
- Neural networks



Complexity

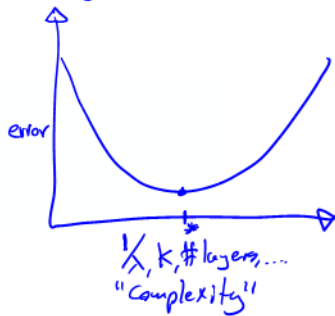
- Depth, leaf purity, ...
- Feature transformation/selection
- # neighbors
- StD. vs NB, α
- # layers, activation functions
- # units

Regularization

$$E(\underline{w}) = \sum_i (y_i - f(x_i; \underline{w}))^2 + \lambda \|\underline{w}\|_2^2$$

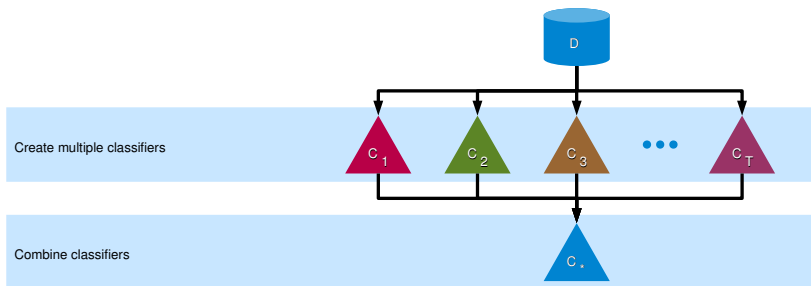
$\uparrow$
!

Two-layer CV



Ensemble methods

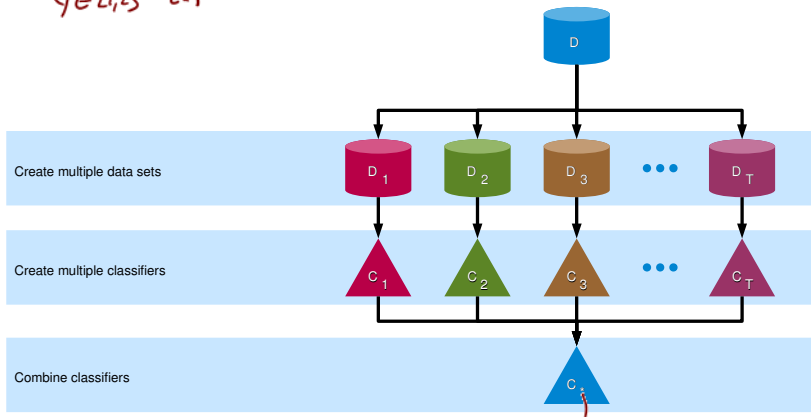
- Combine multiple (weak) classifiers into one (strong) classifier
- Each classifier trained using **different** variations of
 - Data set
 - Input attributes
 - Class label
 - Algorithm/model



Ensemble methods

$$f^*(x_i) = \underset{y \in \{1,2,3\}}{\operatorname{argmax}} \sum_{t=1}^T \delta_{f_t(x_i), y_i}$$

$$\delta_{f_t(x), y} = \begin{cases} 1 & f_t(x) = y \\ 0 & f_t(x) \neq y \end{cases}$$



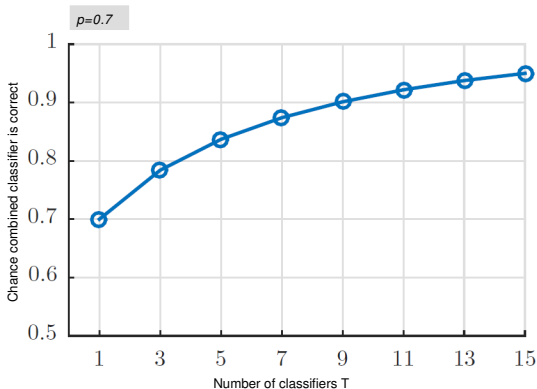
Why ensemble methods?

- Can improve classification algorithms in terms of
 - Better classification accuracy
 - Increased stability
 - Reduced variance
 - Less overfitting
- Consider T **independent classifiers** for binary classification, each with accuracy p .
- The probability a classifier which use majority voting is correct is then given by:

$$\begin{aligned} P(\text{Majority voting is correct}) &= \sum_{t=\lceil T/2 \rceil}^T P(\{t \text{ classifiers are correct}\}) \\ &= \sum_{t=\lceil T/2 \rceil}^T \binom{T}{t} p^t (1-p)^{T-t} \end{aligned}$$

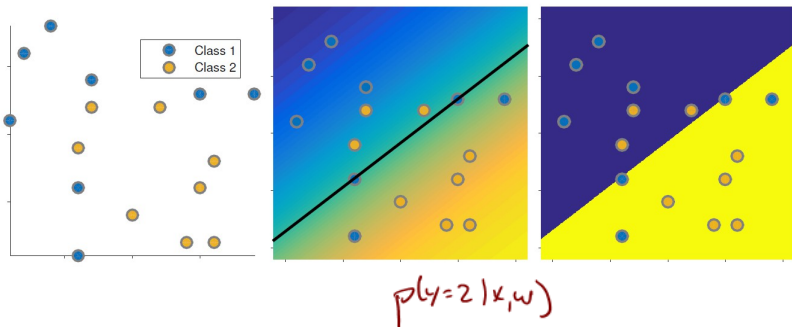
Why ensemble methods?

$$\begin{aligned}P(\text{Majority voting is correct}) &= \sum_{t=\lceil T/2 \rceil}^T P(\{t \text{ classifiers are correct}\}) \\&= \sum_{t=\lceil T/2 \rceil}^T \binom{T}{t} p^t (1-p)^{T-t}\end{aligned}$$



Recall: Logistic regression

- Classification using logistic regression

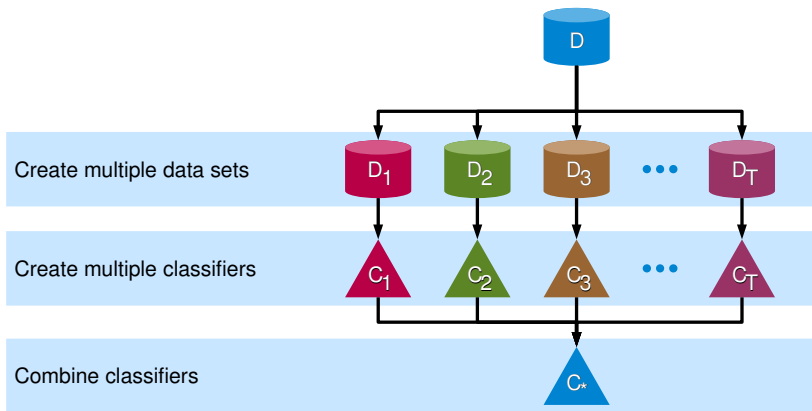


Bagging

- New training data sets drawn randomly from pool with replacement

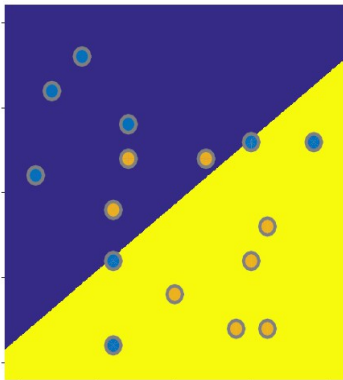
Pool of training data	1	2	3	4	5	6	7	8	9	10
New training data sets	3	5	4	3	9	7	9	5	1	1
	5	8	2	6	2	3	8	3	5	1
	1	7	4	1	10	6	10	8	8	7
	⋮									
	4	3	8	5	2	4	7	10	10	8

Bagging

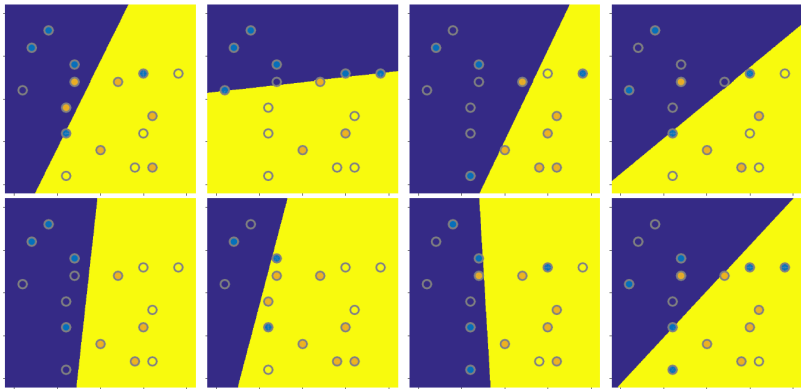


Bagging

- **Single classifier**
 - Logistic regression
 - Two attributes (x_1, x_2)



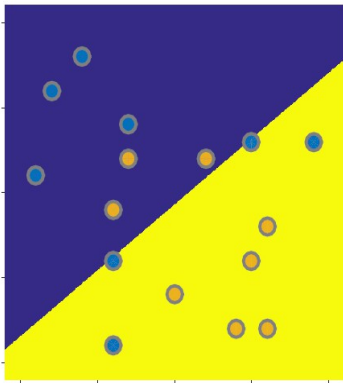
Bagging



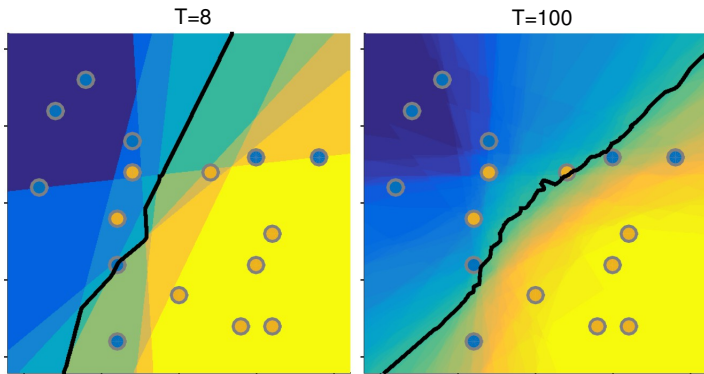
Notice, hollow dots are observations not included in bagging round

Bagging

- Recall: Single classifier



Bagging



Boosting

Pool of training data	1	2	3	4	5	6	7	8	9	10
Weights	.1	.1	.1	.1	.1	.1	.1	.1	.1	.1
New training data set	3	5	4	3	9	7	9	5	1	1
Train classifier										

Boosting

Pool of training data	1	2	3	4	5	6	7	8	9	10
Weights	.1	.1	.1	.1	.1	.1	.1	.1	.1	.1
New training data set	3	5	4	3	9	7	9	5	1	1
Train classifier										
Classify all data objects	1	2 ✓	3	4 ✓	5	6 ✓	7	8	9	10

Boosting

Pool of training data	1	2	3	4	5	6	7	8	9	10
Weights	.1	.1	.1	.1	.1	.1	.1	.1	.1	.1
New training data set	3	5	4	3	9	7	9	5	1	1
Train classifier										
Classify all data objects	1 ✓	2 ✗	3 ✓	4 ✗	5 ✓	6 ✗	7 ✓	8 ✓	9 ✓	10 ✓
Update weights	.07	.17	.07	.17	.07	.17	.07	.07	.07	.07

Boosting

Pool of training data

1	2	3	4	5	6	7	8	9	10
.1	.1	.1	.1	.1	.1	.1	.1	.1	.1

New training data set

3	5	4	3	9	7	9	5	1	1
---	---	---	---	---	---	---	---	---	---

Train classifier



Classify all data objects

1 ✓	2 ✗	3 ✓	4 ✗	5 ✓	6 ✗	7 ✓	8 ✓	9 ✓	10 ✓
.07	.17	.07	.17	.07	.17	.07	.07	.07	.07

Update weights

New training data set

6	4	7	3	2	4	10	2	5	6
---	---	---	---	---	---	----	---	---	---

Train classifier



AdaBoost

Algorithm 6: AdaBoost algorithm

- 1: Initialize $w_i(1) = \frac{1}{N}$ for $i = 1, \dots, N$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: Create $\mathcal{D}_t$ by sampling (with replacement) from $\mathcal{D}$ according to $\mathbf{w}(t)$
- 4: Let f_t be the classifier *trained* on $\mathcal{D}_t$
- 5: $\epsilon_t = \sum_{i=1}^N w_i(t) (1 - \delta_{f_t(\mathbf{x}_i), y_i})$ (*weighted error of f_t on all data*).
- 6: $\alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t}$
- 7: For each i update weights using eq. (15.7):

$$w_i(t+1) = \frac{\tilde{w}_i(t+1)}{\sum_{j=1}^N \tilde{w}_j(t+1)}, \quad \tilde{w}_i(t+1) = \begin{cases} w_i(t)e^{-\alpha_t} & \text{if } f_t(\mathbf{x}_i) = y_i \\ w_i(t)e^{\alpha_t} & \text{if } f_t(\mathbf{x}_i) \neq y_i. \end{cases}$$

- 8: **end for**
 - 9: $f^*(\mathbf{x}) = \arg \max_{y=1,2} \sum_{t=1}^T \alpha_t \delta_{f_t(\mathbf{x}), y}$ (*Majority voting classifier*)
-

AdaBoost

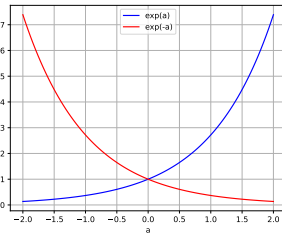
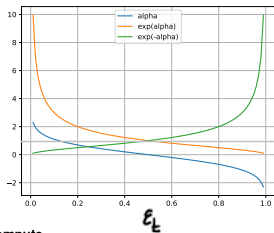
Algorithm 6: AdaBoost algorithm

- 1: Initialize $w_i(1) = \frac{1}{N}$ for $i = 1, \dots, N$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: Create $\mathcal{D}_t$ by sampling (with replacement) from $\mathcal{D}$ according to $\mathbf{w}(t)$
- 4: Let f_t be the classifier trained on $\mathcal{D}_t$
- 5: $\epsilon_t = \sum_{i=1}^N w_i(t) (1 - \delta_{f_t(\mathbf{x}_i), y_i})$ (weighted error of f_t on all data).
- 6: $\alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t}$
- 7: For each i update weights using eq. (15.7):

$$w_i(t+1) = \frac{\tilde{w}_i(t+1)}{\sum_{j=1}^N \tilde{w}_j(t+1)}, \quad \tilde{w}_i(t+1) = \begin{cases} w_i(t)e^{-\alpha_t} & \text{if } f_t(\mathbf{x}_i) = y_i \\ w_i(t)e^{\alpha_t} & \text{if } f_t(\mathbf{x}_i) \neq y_i. \end{cases}$$

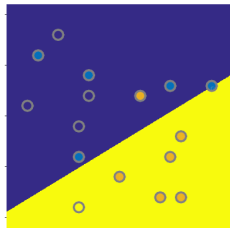
8: **end for**

9: $f^*(\mathbf{x}) = \arg \max_{y=1,2} \sum_{t=1}^T \alpha_t \delta_{f_t(\mathbf{x}), y}$ (Majority voting classifier)

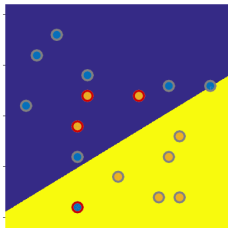


AdaBoost

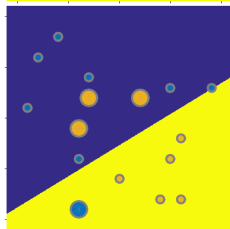
A:
A dataset is sampled with replacement and a classifier trained.



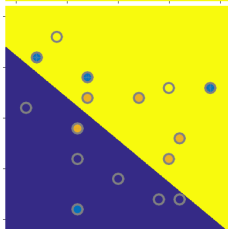
B:
Mis-classified observations are identified.



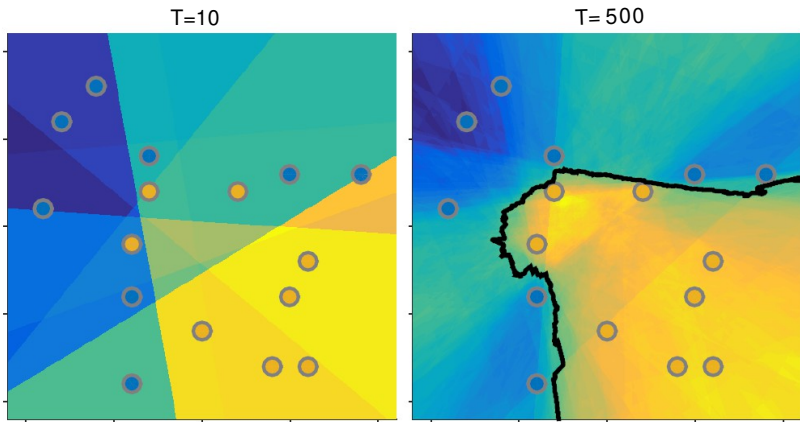
C:
Weights are updated such that more emphasis is given to these mis-classified observations.



New round:
Based on the updated weights a new dataset is sampled and a classifier trained (shown), mis-classified observations identified and given more emphasis...



AdaBoost

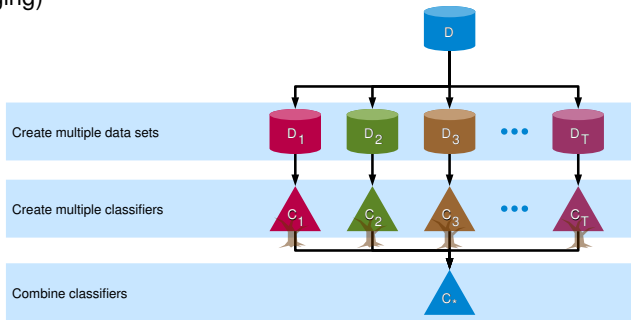


Bagging example: Random forest

Each tree is generated as follows:

- Sample dataset with replacement
- When generating each node in the tree, randomly select a subset of the features and only consider splits using these features

A large number of trees are generated and the trees are combined using majority voting (bagging)



14:55

Quiz 2: Adaboost

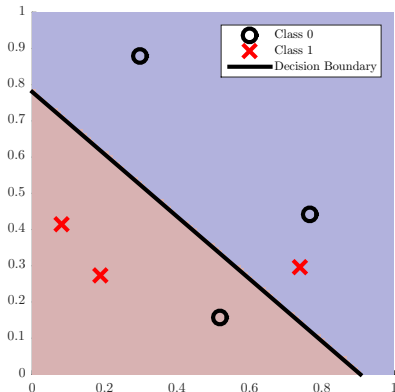


Figure 1: A binary classification problem and the decision boundary obtained by logistic regression. Observations left of the boundary are classified as belonging to the positive class 1 (red crosses) and observations right of the boundary to the negative class 0 (black circles)

$$e^{\ln 5 \sqrt{2}} = 5\sqrt{2}$$

$$e^{-\ln 5 \sqrt{2}} = \frac{1}{5\sqrt{2}}$$

$$\sqrt{2} = \frac{2}{\sqrt{2}}$$

We wish to apply a logistic regression model to the binary classification problem shown in Figure 1. We attempt to improve the performance by applying AdaBoost. AdaBoost works by first sampling a new dataset with replacement, then training a classifier on the dataset and then proceeding with the subsequent steps of the AdaBoost algorithm.

Suppose in the first iteration of the AdaBoost algorithm the classification boundary of the trained classifier is as indicated by the black line (i.e. observations left of the black line are classified as in the positive class). What is the resulting value of the weights w ?

- ☒ A. $w = [0.125 \quad 0.250 \quad 0.125 \quad 0.125 \quad 0.125 \quad 0.250]$
- ☐ B. $w = [0.026 \quad 0.447 \quad 0.026 \quad 0.026 \quad 0.026 \quad 0.447]$
- ☒ C. $w = [0.235 \quad 0.029 \quad 0.235 \quad 0.235 \quad 0.235 \quad 0.029]$
- ☐ D. $w = [0.1 \quad 0.3 \quad 0.1 \quad 0.1 \quad 0.1 \quad 0.3]$
- ☐ E. Don't know.

(Hint: First compute ε_1 , then α_1 , then the weights)

Solution:

$$\varepsilon_t = 2/6 \quad \alpha_t = \frac{1}{2} \log \frac{1 - \varepsilon_t}{\varepsilon_t} = \frac{1}{2} \log \frac{4/6}{2/6} = \frac{1}{2} \log 2 = \log \sqrt{2}$$

Not correct: $\tilde{w}_i(t+1) = w_i(t) e^{\alpha_t} = \frac{1}{6} \cdot e^{\log \sqrt{2}} = \frac{1}{6} \sqrt{2}$

$$w_+(t+1) = \frac{\frac{1}{6} \sqrt{2}}{\frac{4}{6} \sqrt{2}} = \frac{1}{4} = 0.25$$

Correct: $\tilde{w}_i(t+1) = w_i(t) e^{-\alpha_t} = \frac{1}{6} e^{-\log \sqrt{2}} = \frac{1}{6} \frac{1}{\sqrt{2}}$

$$w_-(t+1) = \frac{\frac{1}{6} \frac{1}{\sqrt{2}}}{\frac{4}{6} \sqrt{2}} = \frac{1}{8} = 0.125$$

Normalization: $\sum_i \tilde{w}_i(t+1) = 2 \cdot \frac{1}{6} \sqrt{2} + 4 \cdot \frac{1}{6} \cdot \frac{1}{\sqrt{2}} = \frac{2}{6} \sqrt{2} + \frac{2}{6} \sqrt{2} = \frac{4}{6} \sqrt{2}$

The classifier classifies two out of $N = 6$ observations incorrectly. We therefore have:

$$\varepsilon_i = \left[\sum_{j=1}^N w_j I(\hat{y}_j \neq y_j) \right]$$

$$\alpha_i = \frac{1}{2} \log \left(\frac{1 - \varepsilon_i}{\varepsilon_i} \right)$$

and accordingly $\varepsilon_1 = \frac{1}{N} \times 2 = \frac{1}{3}$. This gives

$$\alpha_1 = \frac{1}{2} \log \frac{1 - \frac{1}{3}}{\frac{1}{3}} = \frac{1}{2} \log 2$$

and so for w we get

$$w \propto [e^{-\alpha_1} \quad e^{\alpha_1} \quad e^{-\alpha_1} \quad e^{-\alpha_1} \quad e^{-\alpha_1} \quad e^{\alpha_1}]$$

$$(\text{vs. } \frac{1}{6} \approx 0.166)$$

Simplifying by moving $\frac{1}{\sqrt{2}}$ outside the vector:

$$w \propto [1 \quad 2 \quad 1 \quad 1 \quad 1 \quad 2]$$

and normalizing:

$$w = \frac{1}{8} [1 \quad 2 \quad 1 \quad 1 \quad 1 \quad 2]$$

accordingly option A is correct.

Summary

- Bias/Variance
 - Regularization
- Ensemble methods
 - Bagging (incl. random forrest)
 - Boosting

Resources

<https://towardsdatascience.com> More in-depth discussion of the Random Forrest algorithm and parameter choices

(<https://towardsdatascience.com/the-random-forest-algorithm-d457d499ffcd>)

<https://www.datacamp.com> Practical use of the random forrest algorithm in python (<https://www.datacamp.com/community/tutorials/random-forests-classifier-python>)

<https://citeseerx.ist.psu.edu> Justification for the AdaBoost algorithm (technical) (<https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.51.9525>)